Time Series Modelling Case Study
Forecasting Gas Price Using ARMA and Prophet Models

Git-hub Link: Time-Series-Modelling-Case-Study

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1. Introduction

Forecasting gas prices is important because changes in fuel costs can affect businesses, households, and wider economic activity. Sudden increases or decreases in energy prices can influence operating costs, consumer spending, and financial planning. As a result, organisations often rely on forecasting techniques to better understand possible future price movements and support decision-making.

Time series forecasting uses historical observations to identify patterns that may continue into the future. Different forecasting models are available, each with strengths and limitations depending on the characteristics of the data. Traditional statistical approaches such as the Autoregressive Moving Average (ARMA) model are widely used when a time series satisfies the assumptions of stationarity. More recently, Prophet has gained popularity as a forecasting tool because of its ability to model changing trends while providing measures of uncertainty.

This study examines a dataset containing 500 daily gas price observations collected between January 2025 and June 2026. The analysis investigates the behaviour of the series, assesses whether transformations are required to achieve stationarity, develops an ARMA forecasting model, evaluates Prophet as an alternative approach, and compares the predictive performance of both methods using commonly used forecasting accuracy measures.

2. Dataset Overview and Exploratory Data Analysis

Before developing forecasting models, the dataset was reviewed to ensure that the information was suitable for analysis. Checks were performed for missing values, duplicate records, duplicated dates, and potential gaps in the daily time sequence. No major quality issues were identified, indicating that the dataset could be used without substantial preprocessing.

	price
count	500.000000
mean	57.876340
std	16.402565
min	25.440000
25%	41.685000
50%	61.295000
75%	69.347500
max	86.660000

Table 1. Summary Statistics of Gas Prices

Summary statistics showed that gas prices varied considerably throughout the observation period. While the average price was approximately 57.9 units, individual observations ranged from around 25 units to almost 87 units. This level of variation suggests that gas prices experienced meaningful fluctuations over time and therefore present an interesting forecasting problem.

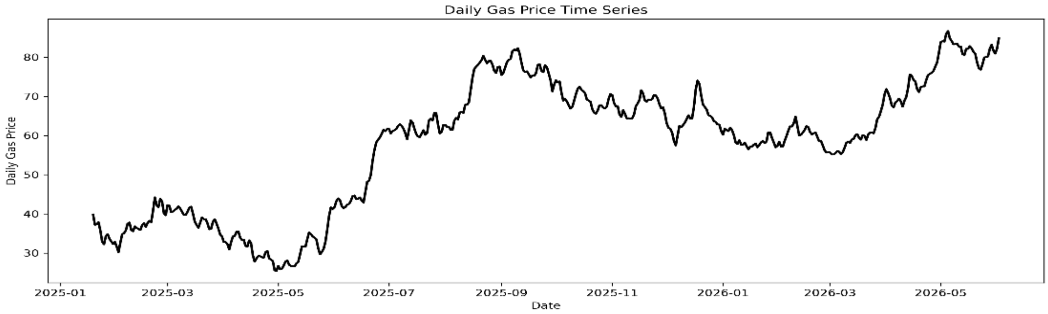

Figure 1. Daily Gas Price Series

Visual inspection of the time series revealed that prices did not remain around a constant level throughout the study period. Instead, periods of growth and decline were observed, with a noticeable increase during the middle portion of the dataset. This behaviour suggests that the underlying characteristics of the series may have changed over time.

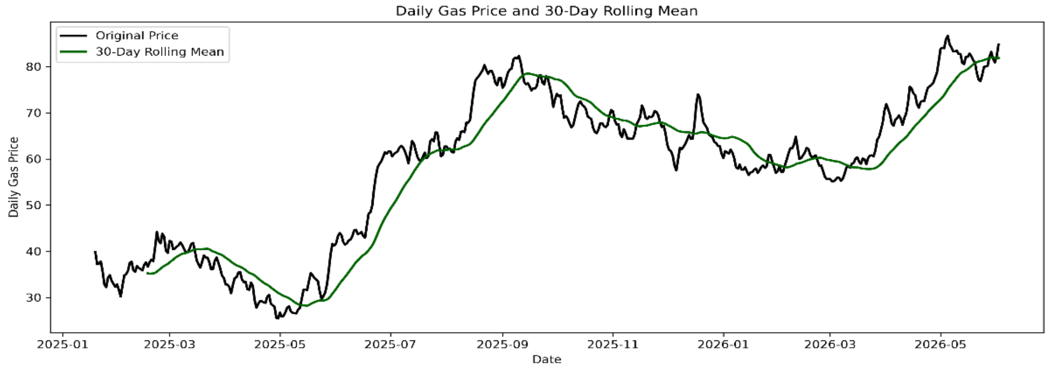

Figure 2. Rolling Mean

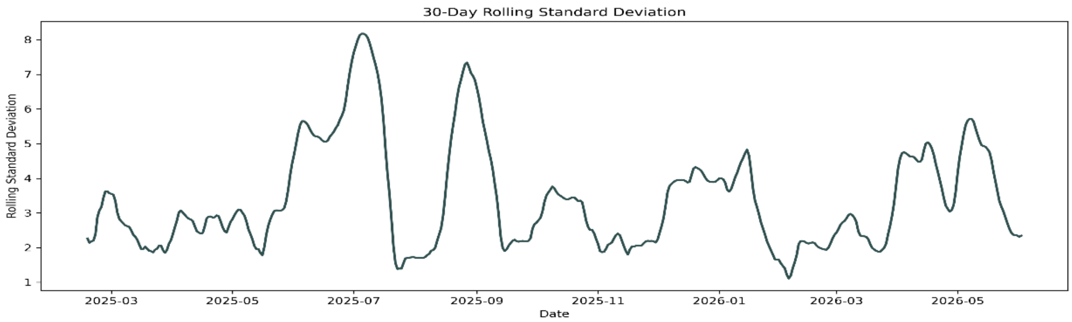

Figure 3. Rolling Standard deviation

To investigate this further, rolling mean and rolling standard deviation statistics were calculated. Both measures showed noticeable variation across the observation period rather than remaining constant. These patterns provided an initial indication that the original gas price series was unlikely to satisfy the stationarity assumption required for ARMA modelling.

3. Stationarity Analysis

Stationarity is an important requirement for ARMA modelling because the statistical properties of the series should remain relatively stable over time. To formally assess stationarity, the Augmented Dickey-Fuller (ADF) test was applied to the original gas price series (Dickey and Fuller, 1979).

Series	ADF Statistic	p-value	Conclusion
Original Series	-0.964	0.766	Non-stationary

Series	ADF Statistic	p-value	Conclusion
First-Differenced Series	-7.112	3.91×10^{-10}	Stationary

Table 2. Augmented Dickey-Fuller Test Results

The results indicated that the original series was non-stationary, as the p-value was substantially higher than the conventional 0.05 significance level. This finding was consistent with the patterns observed during exploratory analysis, where changes in both trend and variability were visible.

To address this issue, first-order differencing was applied. Rather than analysing the absolute price values, differencing focuses on the change between consecutive observations.

The ADF test was repeated after differencing and returned a p-value of 3.91×10^{-10}. This substantial reduction in the p-value indicated that the non-stationarity present in the original series had been successfully removed. As a result, the transformed series was considered suitable for ARMA modelling.

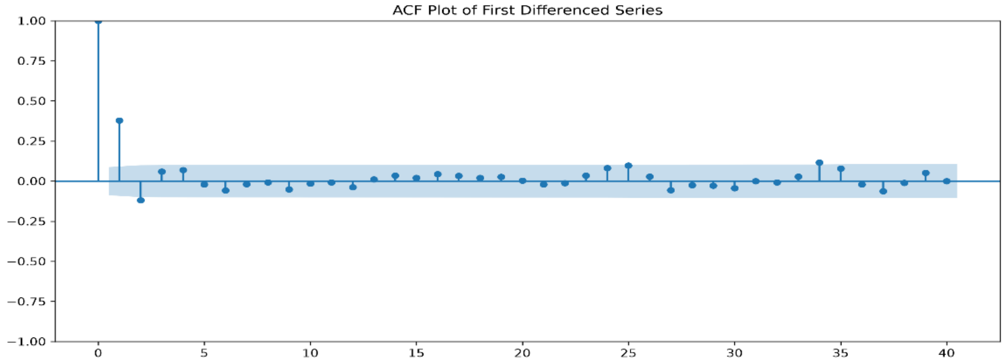

Figure 4. ACF of the Differenced Series

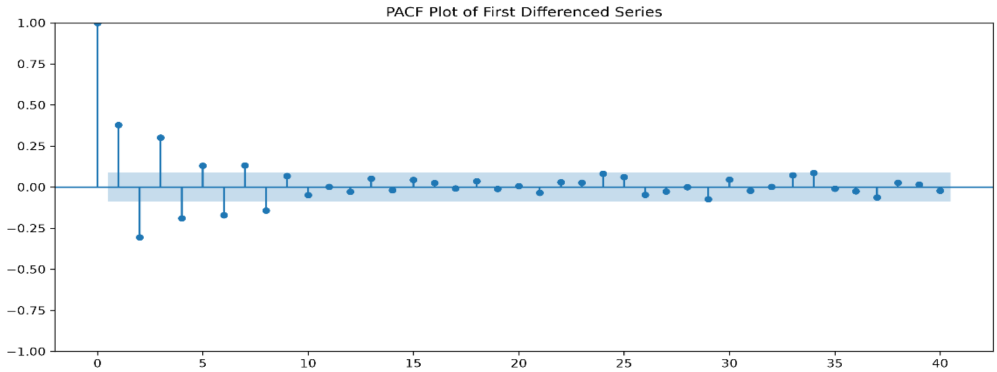

Figure 5. PACF of the Differenced Series

The ACF and PACF plots of the differenced series were then examined to identify remaining autocorrelation patterns. Several early lags remained significant, suggesting that recent observations continued to contain useful forecasting information. Overall, the results indicated that first-order differencing successfully prepared the series for ARMA modelling.

Taken together, the rolling statistics, ADF test results, and autocorrelation analysis showed that first-order differencing successfully transformed the gas price series into a stationary form suitable for ARMA modelling.

4. Forecasting Methodology

A training and testing approach was used to evaluate forecasting performance. The first 440 observations were allocated to model development, while the final 60 observations were reserved for testing. Keeping part of the dataset separate allowed the models to be evaluated on unseen observations, providing a better indication of how they would perform when forecasting future gas prices.

4.1 ARMA Model Development

After confirming that first-order differencing had produced a stationary series, an ARMA model was developed using the transformed data. ARMA was selected because it is well suited to modelling short-term dependencies within stationary time series and has been widely applied in economic and energy forecasting studies (Box et al., 2015).

To identify the most suitable model, a grid search was carried out across different combinations of p, d and q values. Model selection was based on the Akaike Information Criterion (AIC), where lower values indicate a better balance between model fit and complexity (Akaike, 1974). Once stationarity had been achieved through first-order differencing, the final ARMA(5,8) model was fitted to the transformed series.

	p	d	q	AIC
0	5	1	8	1221.652511
1	2	2	8	1223.250971
2	2	1	7	1223.937871
3	5	1	7	1224.604992
4	3	1	7	1224.916640
5	2	2	7	1225.026932
6	5	2	8	1225.070144
7	3	2	8	1225.132594
8	0	1	8	1225.145420
9	3	1	8	1225.638444

Table 3. Best ARMA Model Based on AIC

The search identified (5,1,8) as the optimal specification, producing the lowest AIC value among the candidate models. Since first-order differencing had already been applied to achieve stationarity, the final forecasting model was implemented as an ARMA(5,8) model on the differenced series.

This result suggests that both recent price movements and historical forecast errors contributed useful information for predicting future gas price changes. The selected model was subsequently fitted to the training dataset and used to generate forecasts for the testing period.

4.2 Prophet Model Development

Prophet was implemented as an alternative forecasting approach to compare its performance with the ARMA model. Unlike ARMA, Prophet does not require stationary data and is designed to capture changes in trend while also providing prediction intervals (Taylor and Letham, 2018).

To improve model performance, several values of the changepoint_prior_scale parameter were evaluated. This parameter controls how readily the model responds to changes in trend behaviour.

Lower values produce smoother trend estimates, whereas higher values allow the model to adapt more quickly to changes in the data.

	changepoint_prior_scale	MAE	RMSE	MAPE (%)
0	0.50	14.882736	15.698373	18.746497
1	0.01	16.711107	17.724668	21.017973
2	0.10	17.328698	18.182322	21.853586
3	0.05	18.193794	19.049441	22.957280

Table 4. Prophet Hyperparameter Tuning Results

The hyperparameter tuning results identified a changepoint_prior_scale value of 0.50 as the optimal configuration. This setting produced the lowest forecasting error across the candidate models and was therefore selected for the final Prophet implementation.

4.3 Forecast Evaluation

Forecasting performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). These metrics compare predicted values with observed values, with lower scores indicating better forecasting accuracy. Using multiple evaluation measures provides a broader assessment of model performance because each metric captures forecasting error in a slightly different way (Hyndman and Athanasopoulos, 2021). Together, these measures were used to compare the predictive accuracy of the ARMA and Prophet models on the test dataset

5. Results and Model Comparison

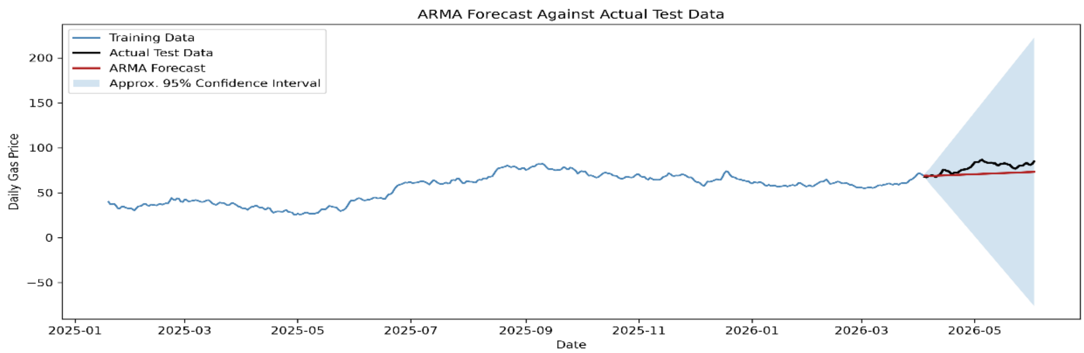

Figure 6. ARMA Forecast Compared with Actual Values

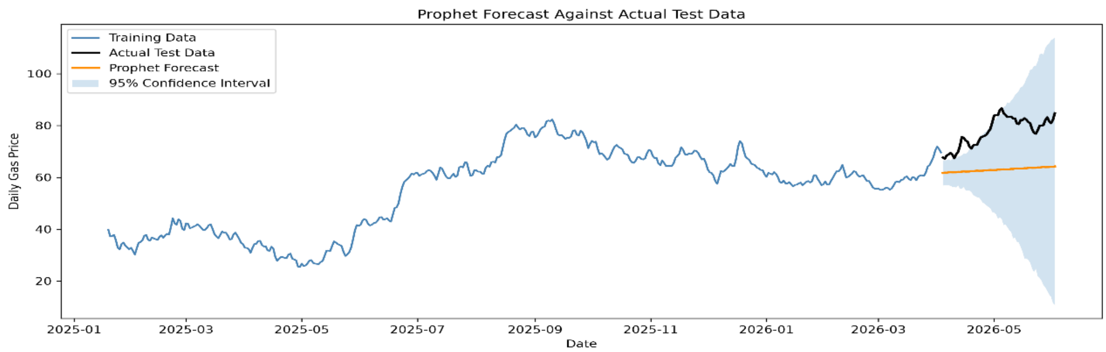

Figure 7. Prophet Forecast Compared with Actual Values

ARMA tracked the observed gas price movements more closely, whereas Prophet produced smoother forecasts that were less responsive to short-term fluctuations. This resulted in better forecasting accuracy for the ARMA model.

	Model	MAE	RMSE	MAPE (%)
0	ARMA on First Differenced Series	7.180442	8.362644	8.880741
1	Prophet	14.882736	15.698373	18.746497

Table 5. Forecast Accuracy Comparison

The forecasting results were consistent with the visual comparison of the models. ARMA achieved lower forecasting errors than Prophet, recording an MAE of 7.18, an RMSE of 8.36, and a MAPE of 8.88%. These results indicate that ARMA provided more accurate predictions during the test period.

The better performance of ARMA is likely related to the way the gas price data behaved after differencing. Short-term patterns were still present in the series, and ARMA was able to capture these changes more effectively than Prophet. As a result, ARMA produced more accurate forecasts during the testing period.

Although Prophet is a flexible forecasting method, its strengths are often more apparent when clear trend changes or strong seasonal patterns exist. In this study, seasonality was relatively weak, which limited the benefits of Prophet's trend-based approach and contributed to larger forecasting errors.

Overall, the results suggest that model performance depends on how well a method matches the characteristics of the data. For this dataset, the ARMA(5,8) model fitted to the first-differenced series produced the most accurate forecasts and was therefore selected as the preferred forecasting model.

6. Residual Diagnostics, Future Forecasting and Limitations

6.1 Residual Diagnostics

Accurate forecasts alone do not guarantee that a model is appropriate. It is also important to examine the residuals to determine whether the model has successfully captured the main patterns present in the data. For this reason, residual diagnostics were conducted for the final ARMA(5,8) model fitted to the first-differenced series.

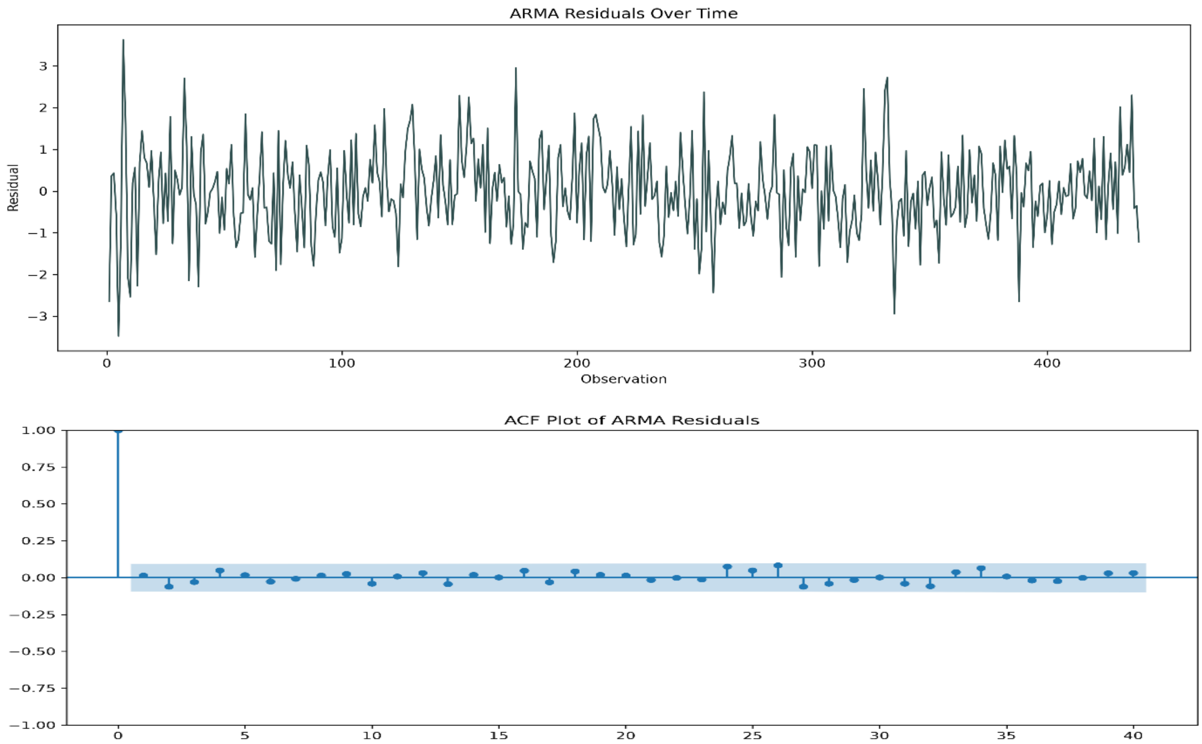

Figure 8. Residual Diagnostics for ARMA(5,8)

The residuals fluctuated around zero with no obvious trend or recurring pattern. This indicates that most of the information contained in the series was captured by the model and that little systematic structure remained unexplained.

The Ljung-Box test was then used to assess whether any autocorrelation remained in the residuals after model fitting (Ljung and Box, 1978).

	lb_stat	lb_pvalue
10	4.820930	0.902814
20	8.951754	0.983459

Table 6. Ljung-Box Test Results

The Ljung-Box test produced p-values greater than 0.05 at both lag 10 and lag 20, indicating no statistically significant evidence of residual autocorrelation. Together with the residual plots and ACF results, this suggests that the ARMA model adequately captured the main temporal patterns present in the differenced series and was suitable for forecasting.

6.2 Future Forecasting

After evaluating model performance, the final models were retrained using the full dataset and used to generate forecasts for the next 24 months.

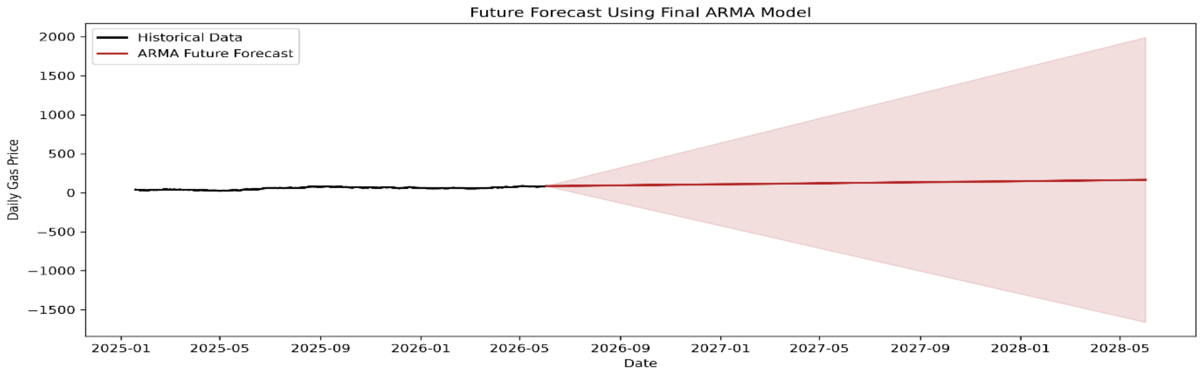

Figure 9. Twenty-Four Month Forecast Generated Using Final ARMA Model

The 24-month forecast suggests that gas prices are likely to remain fairly stable over time. Although the forecast shows a small increase from recent values, the overall change is limited. The confidence intervals become wider as the forecast extends further into the future, indicating greater uncertainty in longer-term predictions.

A key feature of the forecast is the widening confidence intervals over time. This reflects increasing uncertainty as the forecast extends further into the future. While the point forecasts provide an indication of likely future values, the wider intervals show that a broad range of outcomes remains possible.

As gas prices are influenced by many external factors that were not included in this analysis, the forecasts should be viewed as projections based on past trends rather than exact predictions of future prices.

6.3 Limitations

A few limitations should be considered when interpreting these findings. The analysis was based on approximately sixteen months of data, which may not fully capture longer-term changes in gas price behaviour. In addition, both models relied only on historical price information and did not consider external influences such as weather conditions, economic developments, or geopolitical events that can affect energy markets. While ARMA produced the most accurate forecasts for this dataset, its performance may differ when applied to data with stronger trends or seasonal patterns.

Future studies could improve forecasting accuracy by incorporating external variables and evaluating other approaches, such as SARIMA, machine learning, or hybrid forecasting models.

7. Conclusion

This study compared ARMA and Prophet models for forecasting daily gas prices between January 2025 and June 2026. The exploratory analysis and stationarity testing showed that the original series was non-stationary and required first-order differencing before modelling.

Using an AIC-based grid search, the optimal specification was identified as (5,1,8), and the final forecasting model was implemented as ARMA(5,8) on the differenced series. When evaluated on unseen data, ARMA achieved lower forecasting errors than Prophet across all evaluation metrics, demonstrating stronger predictive performance for this dataset.

The findings suggest that short-term patterns in the gas price series played a greater role in forecasting accuracy than trend-based modelling approaches. The future forecasts suggested relatively stable gas price behaviour over the forecast horizon, although the level of uncertainty increased substantially as the prediction period extended further into the future.

Overall, the ARMA(5,8) model provided the most accurate forecasts and was therefore selected as the preferred forecasting approach. Future studies could improve forecasting accuracy by incorporating external variables and exploring more advanced forecasting techniques.

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